

Tests of Significance: Large and Small Samples

Theory: Large Sample Tests ($n > 30$)

- Used when sample size is large ($n > 30$), so the sampling distribution of the mean is approximately normal.
- If population standard deviation (σ) is known, we use **z-test**. If unknown, sample standard deviation is used.
- Tests whether the difference between two sample means is due to chance.
- Hypotheses:

$$H_0 : \mu_1 = \mu_2 \quad (\text{no difference}), \quad H_1 : \mu_1 \neq \mu_2 \quad (\text{difference exists})$$

Two Independent Samples from Same Population

Standard Error of Difference:

$$\text{S.E.}(\bar{X}_1 - \bar{X}_2) = \sqrt{\sigma^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$$

- σ : population standard deviation
- If unknown, substitute pooled sample standard deviation

Two Independent Samples from Different Populations

Standard Error:

$$\text{S.E.}(\bar{X}_1 - \bar{X}_2) = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

- σ_1, σ_2 : population standard deviations
- If unknown, replace with S_1, S_2 (sample standard deviations)

Example: Intelligence Test of Boys and Girls

Data:

Group	Mean	S.D.	Sample Size
Girls	75	15	150
Boys	70	20	250

Step 1: Standard Error

$$\text{S.E.} = \sqrt{\frac{15^2}{150} + \frac{20^2}{250}} = 1.761$$

Step 2: Test Statistic

$$Z = \frac{75 - 70}{1.761} = 2.84$$

Step 3: Conclusion $Z > 2.58$ (1% level) $\Rightarrow$ Reject H_0 . There is a significant difference.

Theory: Small Sample Tests ($n \leq 30$)

- Used when sample size is small ($n \leq 30$) and population standard deviation is unknown.
- Assumes the population is approximately normal.
- Based on **Student's t-distribution** (William Gosset, 1908).
- Test Statistic:

$$t = \frac{\bar{X} - \mu}{S/\sqrt{n}}, \quad \nu = n - 1$$

- Wider tails than normal distribution $\rightarrow$ confidence intervals are wider.

Example: Life of Electric Bulbs

Data: $\mu = 25$, $n=6$, sample: 24, 26, 30, 20, 20, 18

Step 1: Sample Mean

$$\bar{X} = \frac{138}{6} = 23$$

Step 2: Sum of Squares

$$\sum (X_i - \bar{X})^2 = 102$$

Step 3: Sample SD

$$S = \sqrt{\frac{102}{5}} = 4.517$$

Step 4: t-Statistic

$$t = \frac{|23 - 25|}{4.517/\sqrt{6}} = 1.084$$

Step 5: Conclusion $t < t_{0.01} = 4.032 \Rightarrow$ Accept H_0 . Producer's claim is valid.

Two Independent Samples t-Test

Used to test difference between means of two independent samples.

$$t = \frac{\bar{X}_1 - \bar{X}_2}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}, \quad S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$$

Hypotheses: $H_0 : \mu_1 = \mu_2$, $H_1 : \mu_1 \neq \mu_2$

Example: Efficacy of Two Drugs

Data:

Drug A	10	12	13	11	14	
Drug B	8	9	12	14	15	10
	9					

Step 1: Sample Means

$$\bar{X}_1 = 12, \quad \bar{X}_2 = 11$$

Step 2: Sample Variances

$$S_1^2 = 2.5, \quad S_2^2 = 7.33$$

Step 3: Pooled SD

$$S_p \approx 2.345$$

Step 4: t-Statistic

$$t \approx 0.67$$

Conclusion: Not significant. No difference between drugs.

Paired Sample t-Test (Dependent Samples)

- Used when samples are related (before-after, matched pairs).
- Test Statistic:

$$t = \frac{\bar{d}}{S_d/\sqrt{n}}, \quad S_d = \sqrt{\frac{\sum d_i^2 - n\bar{d}^2}{n-1}}$$

- Degrees of freedom: $n - 1$

Example: Weight Before and After Health Club

Participant	Before	After	d=After-Before
1	120	111	-9
2	125	114	-11
3	115	107	-8
4	130	120	-10
5	123	115	-8
6	119	112	-7
7	122	112	-10
8	127	120	-7
9	128	119	-9
10	118	112	-6

$$\bar{d} = -8.5, \quad S_d = 1.58, \quad t = 17$$

Conclusion: Significant weight reduction after joining the club.

Practice Qtn

For a random sample of 10 persons fed on **Diet A**, the increase in weight (in pounds) during a certain period was:

10, 6, 16, 17, 13, 12, 8, 14, 15, 9

7, 23, 13, 22, 15, 12, 14, 18, 8, 21, 10, 17

Test whether the diets A and B differ significantly with respect to their effect on increase in weight.

Given:

Degrees of Freedom	Value of t at 5% level
19	2.09
20	2.09
21	2.08
22	2.07
23	2.07

Practice Question (Data)

Two laboratories **A** and **B** independently estimate the fat content (in grams) of ice cream. Each batch sample is halved and sent to both laboratories.

Batch No.	1	2	3	4	5	6	7	8	9	10
Lab A	7	8	7	3	8	6	9	4	7	8
Lab B	9	8	8	4	7	7	9	6	6	6

Test whether there is a significant difference between the mean fat contents reported by laboratories A and B.

Practice Question (Reference Table)

Use the following extract from the t -table:

Degrees of Freedom	6	7	8	9	10	16	18	20
5% Value of t	1.45	2.36	2.31	2.26	2.23	2.12	2.10	2.09

Practice Question (Data)

In a test given to two groups of students, the marks obtained are as follows:

First Group	18	20	36	50	49	36	34	49	41
Second Group	29	28	26	35	30	44	46		

Examine whether there is a significant difference between the arithmetic mean of marks secured by the two groups.

Practice Question (Given)

The value of t at the 5% level of significance for

$$\nu = 14$$

is given as:

$$t_{0.05} = 2.14$$